Name: _____ Date: _____ Period: _____

Unit 3 Review Packet · Sheet 1 of 3

Step 1: Self-check. Mark one box in each row (be honest; this tells you what to practice tonight).

Obj. (try)	I can …	Got it	Almost	Not yet
3-1 (#1–4)	factor out the GCF, such as $12x + 6 = 6(2x + 1)$, and keep the 1 when the GCF is a whole term.			
3-2 (#5–8)	factor quadratics when the leading coefficient is 1, and when it is greater than 1.			
3-3 (#9–12)	complete the square to write $(x - p)^2 = q$ and solve (even when $a > 1$), and show where the quadratic formula comes from.			
3-4 (#13–16)	connect roots and factors, build a real polynomial from real and complex roots, and explain with the Fundamental Theorem of Algebra.			
3-5 (#17–20)	recognize a difference of squares and factor it completely.			
3-6 (#21–24)	recognize sums and differences of cubes and factor them correctly.			

Step 2: Practice. Show your work, and check each factorization by multiplying back. 3-1: Greatest common factor

1. (B) Factor out the GCF: $14x + 21$

2. (B) Factor out the GCF: $9x^2 + 3x$

3. (B) Factor out the GCF: $12x^4 - 18x^3 + 6x^2$

4. (B) Solve by factoring out the GCF: $6x^2 - 15x = 0$

3-2: Factoring quadratics

5. (B) Factor: $x^2 + 9x + 20$

6. (B) Factor: $x^2 - 3x - 28$

7. (I) Factor completely: $2x^2 - 14x + 20$

8. (I) Solve by factoring: $5x^2 - 13x - 6 = 0$

Name: _____ Date: _____ Period: _____

Unit 3 Review Packet · Sheet 2 of 3 · (B = Basic, I = Intermediate, A = Advanced)

3-3: Completing the square

9. (B) Complete the square to rewrite $x^2 - 6x + 5 = 0$ as $(x - p)^2 = q$. Then solve.

10. (B) Solve by completing the square: $x^2 - 10x + 18 = 0$. Give exact answers.

11. (I) Solve by completing the square: $2x^2 - 8x + 5 = 0$. Give exact answers.

12. (A) **Derive the quadratic formula.** Start with $ax^2 + bx + c = 0$ ($a \neq 0$). (1) Divide by a. (2) Move the constant to the right. (3) Add $(b/2a)^2$ to *both* sides. (4) Factor the left side and combine the right side. (5) Take square roots. (6) Solve for x. Then use your formula to solve $x^2 - 4x + 13 = 0$.

3-4: Fundamental Theorem of Algebra

13. (B) $f(x) = (x - 3)^2(x + 4)(x^2 + 1)$. Give the degree and list every root, real and complex, with its multiplicity.

14. (B) A degree-4 polynomial with real coefficients has roots 2, −1, and 3 + i. What is the fourth root? Write f in factored form.

15. (I) Write the lowest-degree polynomial with real coefficients and roots −2 and 1 + 2i. Give it in standard form.

16. (A) Maya says a degree-5 polynomial with real coefficients could have no real roots. (a) Use the theorem and conjugate pairs to explain why she is wrong. (b) If its graph crosses the x-axis only at $x = -2$ (a simple root), describe the other four roots.

Name: _____ Date: _____ Period: _____

3-5: Difference of squares

17. (B) Factor: $4x^2 - 81$

18. (B) Which are a difference of squares? Factor those that are and say why the others are not. (a) $x^2 + 16$ (b) $49x^2 - 4$ (c) $x^2 - 18$ (d) $100 - y^2$

19. (B) Factor completely: $x^4 - 16$

20. (B) Factor completely: $3x^2 - 75$

3-6: Sum and difference of cubes

21. (B) Label each as a sum of cubes, a difference of cubes, or neither. (a) $x^3 + 64$ (b) $27x^3 - 1$ (c) $x^3 + 36$ (d) $125 - x^3$

22. (I) Factor: $x^3 - 125$

23. (I) Factor: $64x^3 + 1$

24. (I) Factor completely: $2x^3 - 54$

Plan for tonight

The two objectives I will practice tonight: _____ and _____.
One trap I will watch for on the test: _____

Math Time Machine: As the story is told, the mathematician Hardy called the taxi number 1729 dull, and Ramanujan replied that it is the smallest number that is a sum of two positive cubes in two different ways: $1^3 + 12^3 = 9^3 + 10^3$. A sum of cubes factors as $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$.